A coupling for prime factors

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Joint work with Dimitris Koukoulopoulos

Prime factors on a logarithmic scale

- Pick an integer N **uniformly** in $[1, x]$.
- Let $P_1 P_2 \cdots$ be the **unique factorization** of N with $P_1 \geqslant P_2 \geqslant \cdots$ being all primes or ones.

Theorem (Billinglsey, 1972)

For any fixed $r \geqslant 1$, the random vector

$$\left(\frac{\log P_1}{\log x}, \ldots, \frac{\log P_r}{\log x}\right) \xrightarrow{d} (V_1, \ldots, V_r)$$

where $\mathbf{V} := (V_1, V_2, \ldots)$ satisfy a **Poisson-Dirichlet distribution**.

Poisson-Dirichlet distribution

Defining it from a stick-breaking process:

- Sample $(U_i)_{i \geqslant 1}$ independently and uniformly in $[0, 1]$.
- Define
  - $L_1 := U_1$;
  - $L_i := U_i(1 - U_{i-1}) \cdots (1 - U_1)$ for $i \geqslant 2$.

  GEM distribution : Distribution of $\mathbf{L} = (L_1, L_2, \ldots)$.
- Let $\mathbf{V} = (V_1, V_2, \ldots)$ be the non-increasing order statistics of $\mathbf{L}$.
  Poisson-Dirichlet distribution : Distribution of $\mathbf{V} = (V_1, V_2, \ldots)$.

History

- (Billingsley, 1972)
  $\mathbb{P}\left[\frac{\log P_i}{\log x} < u_i \ \forall i \leqslant k\right] = \mathbb{P}\left[V_i < u_i \ \forall i \leqslant k\right] + o_{x\to\infty}(1)$,
- (Tenenbaum, 2000)
  $\mathbb{P}\left[\frac{\log P_i}{\log x} < u_i \ \forall i \leqslant k\right] = \mathbb{P}\left[V_i < u_i \ \forall i \leqslant k\right] + O\left(\frac{(\log 1/u_k)^{k-2}}{\log x}\right)$,
- (Arratia, 2002)
  There exists a **coupling** of N and $\mathbf{V}$ such that
  $\mathbb{E}\sum_{i\geqslant 1}\left|\frac{\log P_i}{\log x} - V_i\right| \ll \frac{\log\log x}{\log x}$.

Definition

A **coupling** of two random variables X and Y is a single probability space where copies of both these random variables live on.

Prime factorizations and cycle decompositions

The **cycle decomposition** of a random permutation $\sigma \in S_n$ &
the **prime factorization** of a random integer $N \in \mathbb{Z} \cap [1, x]$
have similar stats with $n \approx \log x$.

Examples:

	Integers in $[1, x]$	Permtations in S_n
Avg. # components	$\frac{1}{x} \sum_{n \leqslant x} \omega(n) \approx \operatorname{loglog} x$	$\frac{1}{n!} \sum_{\sigma \in S_n} \omega(\sigma) = H_n \approx \log n$
Prob. of one component	$\frac{\pi(x)}{x} \approx \frac{1}{\log x}$ (PNT)	$\frac{\#\{\text{cycles in } S_n\}}{n!} = \frac{1}{n}$

Cycles in random permutations

Pick a random permutation $\sigma \in S_n$ uniformly.

Let $C_i(\sigma)$ be the i^{th} largest cycle of σ.

History

- (Vershik & Shmidt, 1977)
  $\mathbb{P}\left[\frac{C_i(\sigma)}{n} < u_i \ \forall i \leqslant k\right] = \mathbb{P}\left[V_i < u_i \ \forall i \leqslant k\right] + o_{n\to\infty}(1)$,
- (Arratia, Barbour & Tavaré, 2006)
  There exists a **coupling** of σ and $\mathbf{V}$ such that
  $\mathbb{E}\sum_{i\geqslant 1}\left|\frac{C_i(\sigma)}{n} - V_i\right| \sim \frac{\log n}{4n}$.
- They also proved that the expectation above is $\geqslant \frac{\log n}{4n}(1 - o(1))$ for **all couplings**.

Expected ℓ^1-distance

Theorem (Arratia, 2002)

There exists a coupling of N and $\mathbf{V}$ satisfying

$$\mathbb{E}\sum_{i\geqslant 1}\left|\frac{\log P_i}{\log x} - V_i\right| \ll \frac{\log\log x}{\log x}.$$

He conjectured that there exists a coupling such that this expectation is $O(\frac{1}{\log x})$.

Uniform Lower bound for all couplings

Permutations:

$$\mathbb{E}\sum_{i\geqslant 1}|\tfrac{C_i}{n}-V_i| \geqslant \frac{1}{n}\cdot\mathbb{E}\sum_{i\geqslant 1} d(nV_i,\mathbb{Z}) = \frac{1}{n}\int_0^n \frac{d(t,\mathbb{Z})}{t}\mathrm{d}t \asymp \frac{\log n}{n}$$

because $d(t,\mathbb{Z})\asymp 1$ for a large portion of the integral.

Integers:

$$\mathbb{E}\sum_{i\geqslant 1}|\tfrac{\log P_i}{\log x}-V_i| \geqslant \frac{1}{\log x}\cdot\mathbb{E}\sum_{i\geqslant 1} d((\log x)V_i,\mathbb{Z}) = \frac{1}{\log x}\int_0^{\log x} \frac{d(t,\log\mathcal{P})}{t}\mathrm{d}t \asymp \frac{1}{\log x},$$

because there is always a $\log p$ in any interval $[t, t+e^{-0.425t}]$ for t large enough (Baker, Harman & Pintz, 2001).

Expected ℓ^1-distance

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Expected ℓ^1-distance

Theorem (H. & Koukoulopoulos, 2024)

There exists a coupling of N and $\mathbf{V}$ satisfying

$$\mathbb{E}\sum_{i\geqslant 1}\left|\frac{\log P_i}{\log x}-V_i\right| \asymp \frac{1}{\log x}.$$

1. Arratia's coupling refined

2. The Coupling Method: Random Factorizations

Arratia's original coupling

Sample $\mathbf{L} = (L_1, L_2, \ldots)$ having a GEM distribution.

Extracting a Poisson-Dirichlet distribution:

- Let $(V_i)_{i \geqslant 1}$ be the decreasing order statistics of the L_i's.

Extracting a random integer:

- Approximate x^{L_i} by a close prime power (or one) Q_i deterministically.
- Set $J := \prod_{i \geqslant 2} Q_i$.
- Pick $P_{\mathtt{extra}}$ randomly with uniform distribution in $\{\text{primes}\} \cup \{1\}$.
- Set $M := J \cdot P_{\mathtt{extra}}$.

Possible to choose Q_i's such that
$\mathbb{P}[M \in A] = \frac{|A|}{\lfloor x \rfloor} + O\left(\frac{\log \log x}{\log x}\right)$ for any $A \subseteq \mathbb{N} \cap [1, x]$.

Modifying Arratia's coupling

Arratia's original integer:

- Approximate x^{L_i} by a close prime power (or one) Q_i deterministically.
- Set $J := \prod_{i \geqslant 2} Q_i$.
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Possible to choose Q_i's such that

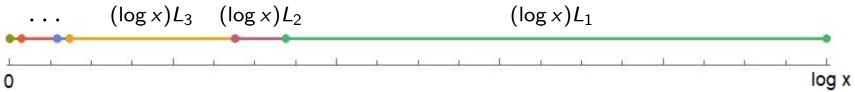
$$\mathbb{P}[M \in A] = \frac{|A|}{\lfloor x \rfloor} + O\left(\frac{\log\log x}{\log x}\right) \quad \forall\, A \subseteq \mathbb{N} \cap [1, x].$$

Our modification:

- Approximate x^{L_i} by a close prime power (or one) Q_i deterministically.
- Set $J := \prod_{i \geqslant 2} Q_i$.
- Pick $P_{\texttt{extra}}$ randomly with $\mathbb{P}[P_{\texttt{extra}} = p \mid J = j] = \frac{\log p}{\theta(x/j)}$.
- Set $M := J \cdot P_{\texttt{extra}}$.

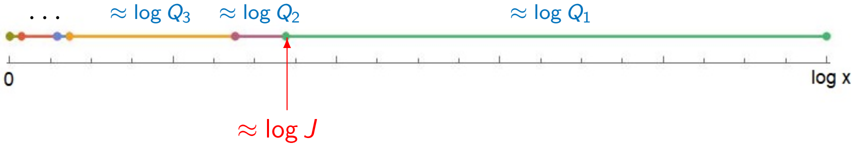
Possible to choose Q_i's such that
$$\mathbb{P}[M \in A] = \frac{|A|}{\lfloor x \rfloor} + O\left(\frac{1}{\log x}\right) \quad \forall\, A \subseteq \mathbb{N} \cap [1, x].$$

Constructing M



- For fixed d, we have $\mathbb{P}[d|J] \to \frac{1}{d}$ as $x \to \infty$. ✓
- But J is **far** from uniform ($\mathbb{P}[J \leqslant y] \approx \frac{\log y}{\log x}$) ✗
- So we need to insert a minimal number of extra primes to **adjust** the distribution. This is why we need an extra prime!

Constructing M



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The **total variation distance** between the laws of two real-valued random variables X and Y is

$$d_{\mathrm{TV}}(X, Y) := \sup_{A \subseteq \mathbb{R}} |\mathbb{P}[X \in A] - \mathbb{P}[Y \in A]|.$$

An equivalent definition is

$$d_{\mathrm{TV}}(X, Y) := \min \mathbb{P}[X \neq Y]$$

where the minimum is over all possible couplings of X and Y.

Recall, that we have $\mathbb{P}[M \in A] = \mathbb{P}[N \in A] + O(\frac{1}{\log x})$,

This implies we can construct a random integer M from N so that $\mathbb{P}[M \neq N] \ll \frac{1}{\log x}$.

When $M = N$

Lemma (H. & Koukoulopoulos, 2024)

1. $\mathbb{P}[M \neq N] \ll \frac{1}{\log x}$.
2. *When $M = N$, then inside the coupling we have*

$$\sum_{i \geqslant 1} \left| \frac{\log P_i}{\log x} - V_i \right| \leqslant \frac{2 \cdot (\log(x/N^\flat) + \Theta_x)}{\log x}$$

with $N^\flat := \prod_{p : v_p(N) = 1} p$ *and* $\Theta_x := \sum_{i \geqslant 1} |\log Q_i / x^{L_i}|$.

What to remember from this:

- For any fixed $\alpha < 1/4$, $\mathbb{E}[(x/N^\flat)^\alpha] + \mathbb{E}[e^{\alpha \Theta_x}] \ll_\alpha 1$.
- The bound when $M = N$ is decomposed
  "arithmetic part" + "probabilistic part".

EXPECTED ℓ^1-DISTANCE

Proof of Theorem:

$$\mathbb{E}\sum_{i\geqslant 1}\left|\frac{\log P_i}{\log x}-V_i\right| \leqslant \mathbb{E}\left[\mathbf{1}_{M=N}\cdot\sum_{i\geqslant 1}\left|\frac{\log P_i}{\log x}-V_i\right|\right]+2\cdot\mathbb{P}[M\neq N]$$

$$\leqslant 2\cdot\frac{\mathbb{E}[\log(x/N^\flat)]+\mathbb{E}[\Theta_x]}{\log x}+2\cdot\mathbb{P}[M\neq N]$$

$$\ll \frac{1}{\log x}.$$

1. Arratia's coupling refined

2. The Coupling Method: Random Factorizations

Extracting Info about divisors

Corollary

If $f : \ell^1(\mathbb{R}) \to \mathbb{R}$ is a K-Lipschitz function, then

$$\mathbb{E}\Big[f\big(\tfrac{\log P_1}{\log x}, \tfrac{\log P_2}{\log x}, \dots\big)\Big] = \mathbb{E}\Big[f\big(V_1, V_2, \dots\big)\Big] + O\Big(\frac{K}{\log x}\Big)$$

For example, this leads to

$$\sum_{n \leqslant x} \log \rho(n) = c \cdot x \log x + O(x).$$

with $\rho(n)$ being the smallest divisor of n larger than $\sqrt{n}$.

Proof:

$$\sum_{n\leqslant x}\log\rho(n)=(\log x)\sum_{n\leqslant x}\frac{\log\rho(n)}{\log n}+O(x).$$

Let Δ^∞ be the sequences of $(v_i)_{i\geqslant 1}$ of positive, decreasing sequences with $\sum v_i = 1$.

For $\mathbf{v}\in\Delta^\infty$, let $\mathcal{D}(\mathbf{v}) := \{\sum_{i\in B} v_i : B\subseteq\mathbb{N}\}$ and let $f(\mathbf{v}) := \min(\mathcal{D}(\mathbf{v})\cap[1/2,1])$.

Then f is 1-Lipschitz on Δ^∞ and

$$\frac{1}{\lfloor x\rfloor}\sum_{n\leqslant x}\frac{\log\rho(n)}{\log n}=\mathbb{E}[f(\tfrac{\log P_1}{\log N},\tfrac{\log P_2}{\log N},\ldots)]=\mathbb{E}[f(\mathbf{V})]+O(\frac{1}{\log x})$$

implying

$$\sum_{n\leqslant x}\log\rho(n)=\mathbb{E}[f(\mathbf{V})]\cdot x\log x+O(x).$$

Here are the **divisors** on a **logarithmic scale** of two distinct integers around 3.52155×10^{25} with exactly 32 divisors:

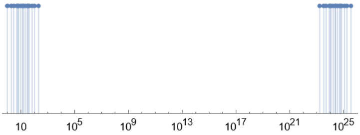

FIGURE: Divisors of $2 \times 3 \times 5 \times 7 \times 16769308972869121317267 1$

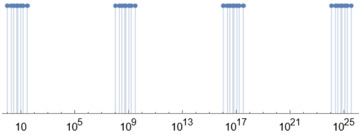

FIGURE: Divisors of $2 \times 3 \times 5 \times 105488303 \times 11127787008134317$

- Let N be a random integer uniform in $\mathbb{N} \cap [1, x]$.
- Let D be a random divisor of N uniformly distributed.

Theorem

For $x \geqslant 2$ and uniformly for $u \in [0, 1]$, we have

$$\mathbb{P}[D < N^u] = \frac{2}{\pi} \arcsin(\sqrt{u}) + \cdots$$

- *Deshouillers, Dress, Tenenbaum (1979):* $\cdots + O\left(\frac{1}{\sqrt{\log x}}\right)$ *for* $u \in [0, 1]$.
- *Bareikis, Manstavičius (2007):* $\cdots + O\left(\frac{1}{\sqrt{u(1-u)}\log x}\right)$ *for* $u \in [\frac{1}{\log x}, 1 - \frac{1}{\log x}]$.

Multiplicative Factorizations into k parts

- *Bareikis, Manstavičius (2007):*

  Estimated $\mathbb{P}[D < N^u]$ when $\mathbb{P}[D = d \mid N = n] = f(d)$ for any $d|n$, and f being **multiplicative**, **fixed at primes**.

- *Nyandwi, Smati (2013):*

  Estimated $\mathbb{P}[D_1 < N^{u_1}$ and $D_2 < N^{u_2}]$ when $D_1 D_2 D_3 = N$ is random 3-factorization (unif. dist.).

- *de la Bretèche, Tenenbaum (2016):*

  Estimated $\mathbb{P}[D_1 < N^{u_1}$ and $D_2 < N^{u_2}]$ when D_1 is a random divisor (unif. dist.), and D_2 is a random divisor of $\frac{N}{D_1}$ (unif. dist.).

- *Leung (2023):*

  Estimated $\mathbb{P}[D_1 < N^{u_1}, \ldots, D_{k-1} < N^{u_{k-1}}]$ for any k, for $\mathbb{P}[D_1 = d_1, \ldots, D_k = d_k \mid N = n] = f(d_1, \ldots, d_k)$ if $d_1 \cdots d_k = n$, and f is **multiplicative**, **fixed at primes**.

The proofs of these theorems all involve getting asymptotics for sums of multiplicative functions (Landau–Selberg–Delange method).

Q: Can we use the distribution of large prime factors to get a similar result?

Probabilistic version:

Theorem (Donnelly and Tavaré, 1987)

- χ *random* k*-colouring of* $\mathbb{N}$*, with* $\mathbb{P}[\chi(i) = j] = \frac{1}{k}$.
- $Z_m := \sum_{i \geqslant 1} 1_{\chi(i)=j} V_i$ *for* $1 \leqslant m \leqslant k$.

Then $\mathbf{Z} := (Z_1, \ldots, Z_k) \sim \mathrm{Dir}(\frac{1}{k}, \ldots, \frac{1}{k})$.

Arratia (1998): Used probabilistic methods to prove
$\mathbb{P}[D < N^u] = \frac{2}{\pi} \arcsin \sqrt{u} + o(1)$.

Since V_i is a model for $\frac{\log P_i}{\log x}$, it is like assigning a random colour to each prime factor to create a random divisor.

Factorizations into k parts

Theorem (Leung, 2023)

Let $\mathbf{u} \in [0,1]^{k-1}$ with $u_1 + \cdots + u_{k-1} \leqslant 1$.
Pick N uniformly in $[1,x]$. Pick $D_1 \cdots D_k = N$ uniformly among all k-factorizations, then

$$\mathbb{P}[D_1 < N^{u_1}, \ldots, D_{k-1} < N^{u_{k-1}}] = F(u_1, \ldots, u_{k-1}) + O_k\left(\frac{1}{(\log x)^{1/k}}\right).$$

Used variant of Landau-Selberg-Delange method.

Factorizations into k parts

Theorem (H., Koukoulopoulos, 2024)

Let $\mathbf{u} \in [0,1]^{k-1}$.
Pick N *uniformly in* $[1,x]$. *Pick* $D_1 \cdots D_k = N$ *uniformly among all* k*-factorizations, then*

$$\mathbb{P}[D_1 < N^{u_1}, \dots, D_{k-1} < N^{u_{k-1}}]$$
$$= F(u_1, \dots, u_{k-1}) + O\left(\frac{1}{\log x} \sum_{1 \leqslant i < k} \frac{1}{(u_i + \frac{1}{\log x})^{1-1/k}(1 - u_i + \frac{1}{\log x})^{1/k}} \right).$$

Error term $\approx \mathbb{P}\left[|Z_i - u_i| < \frac{1}{\log x} \text{ for some } i \right]$.

Coupling Method

Let $\boldsymbol{\delta} = (\frac{\log D_1}{\log N}, \ldots, \frac{\log D_k}{\log N})$ and let $\mathbf{Z}$ be a $\mathrm{Dir}(\frac{1}{k}, \ldots, \frac{1}{k})$ vector.

We want to give a bound on $|\mathbb{P}[\boldsymbol{\delta} \in A] - \mathbb{P}[\mathbf{Z} \in A]|$.

Strategy: Create a coupling of $\boldsymbol{\delta}$ and $\mathbf{Z}$ such that they are close most of the time.

When $M = N$

Lemma (H. & Koukoulopoulos, 2024)

1. $\mathbb{P}[M \neq N] \ll \frac{1}{\log x}$.
2. *When $M = N$, then inside the coupling we have*

$$\sum_{i \geqslant 1} \left| \frac{\log P_i}{\log x} - V_i \right| \leqslant \frac{2 \cdot (\log(x/N^\flat) + \Theta_x)}{\log x}$$

with $N^\flat := \prod_{p: v_p(N)=1} p$ and $\Theta_x := \sum_{i \geqslant 1} |\log Q_i/x^{L_i}|$.

What to remember from this:

- For any fixed $\alpha < 1/4$, $\mathbb{E}[(x/N^\flat)^\alpha] + \mathbb{E}[e^{\alpha \Theta_x}] \ll_\alpha 1$.
- The bound when $M = N$ is decomposed
  "arithmetic part" + "probabilistic part".

When $M = N$

Lemma (H. & Koukoulopoulos, 2024)

1. $\mathbb{P}[M \neq N] \ll \frac{1}{\log x}$.
2. *When $M = N$, then inside the coupling we have*

$$\|\boldsymbol{\delta} - \mathbf{Z}\|_\infty \leqslant \frac{3 \cdot (\log(x/N^\flat) + \Theta_x)}{\log x}$$

with $N^\flat := \prod_{p : v_p(N) = 1} p$ *and* $\Theta_x := \sum_{i \geqslant 1} |\log Q_i / x^{L_i}|$.

Therefore $\mathbb{P}[\boldsymbol{\delta} \in A] = \mathbb{P}[\mathbf{Z} \in A] + O(\frac{1}{\log x} + \mathbb{P}[\mathcal{E}_{\text{arith.}}] + \mathbb{P}[\mathcal{E}_{\text{prob.}}])$ with $\mathcal{E}_{\text{arith.}}$ being the event $B(\boldsymbol{\delta}, \frac{3\log(x/N^\flat)}{\log x})$ intersects ∂A and with $\mathcal{E}_{\text{prob.}}$ being the event $B(\mathbf{Z}, \frac{3\Theta_x}{\log x})$ intersects ∂A.

- Sizes of the prime factors of N converge to components of $\mathbf{V}$

- Construct a space with both N and $\mathbf{V}$, such that prime factors of N are close to components of $\mathbf{V}$ most of the time

- Use this space as a bridge to pass from probabilistic model to get to information on stats of divisors!